

Lab1: Gradient Descend Implementation

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1 Introduction

This report represents the code and the flow for the lab work and the logic implementation to the code of this lab work.

2 Gradient Descend

Gradient Descent is an iterative optimization algorithm used to minimize a cost function by adjusting the model parameters in the direction of the steepest descent of the function's gradient. In simple terms, it finds the optimal values of weights and biases by gradually reducing the error between predicted and actual output.

Cost function:
$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m [(\theta_0 + \theta_1 x^{(i)}) - y^{(i)}]^2$$

Partial derivative w.r.t. θ_0 :
$$\frac{\partial J(\theta_0, \theta_1)}{\partial \theta_0} = \frac{1}{m} \sum_{i=1}^m ((\theta_0 + \theta_1 x^{(i)}) - y^{(i)})$$

Partial derivative w.r.t. θ_1 :
$$\frac{\partial J(\theta_0, \theta_1)}{\partial \theta_1} = \frac{1}{m} \sum_{i=1}^m ((\theta_0 + \theta_1 x^{(i)}) - y^{(i)}) x^{(i)}$$

3 Implementation

The code implemented by the following idea:

Gradient descend

- An **iterative** algorithm to find minimum value
- Idea: use derivative to find the minimum value of a function $f(x)$
 1. Random initialization: $x = x_0$
 2. Assign: $x = x - r * f'(x)$
 3. Re-compute $f(x)$. Stop if $f(x)$ is small enough, or repeat step 2 if not¹

Where $r \geq 0$ is called the *learning rate*.

¹can be many times!

Figure 1: Gradient Descend Logic Flow

The function is design within 3 parameters:

- Learning rate
- Threshold value
- Initial x value

The output is shown as:

```
C:\USTH\USTH_Master_coding\BC4.03_Deep_Learning\dl2026>python lab1.py
Step      | x          | f(x)
-----
0          | 10.0000    | 100.0000
1          | 8.0000     | 64.0000
2          | 6.4000     | 40.9600
3          | 5.1200     | 26.2144
4          | 4.0960     | 16.7772
5          | 3.2768     | 10.7374
6          | 2.6214     | 6.8719
7          | 2.0972     | 4.3980
8          | 1.6777     | 2.8147
9          | 1.3422     | 1.8014
10         | 1.0737     | 1.1529
11         | 0.8590     | 0.7379
12         | 0.6872     | 0.4722
13         | 0.5498     | 0.3022
14         | 0.4398     | 0.1934
15         | 0.3518     | 0.1238
16         | 0.2815     | 0.0792
17         | 0.2252     | 0.0507
18         | 0.1801     | 0.0325
19         | 0.1441     | 0.0208
20         | 0.1153     | 0.0133
21         | 0.0922     | 0.0085
22         | 0.0738     | 0.0054
23         | 0.0590     | 0.0035
```

Figure 2: Gradient Descend Logic Flow